Local-Unitary Rigidity and Transversal Clifford Groups for MDS–CSS AME States

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Abstract

Let C be a linear $[2m, m, m+1]_q$ MDS code, and let $|\Psi_C\rangle = q^{-m/2} \sum_{c\in C}|c\rangle$ be its equal-phase CSS state. For every prime power q and $m \geq 2$, every product-unitary intertwiner between two such states is local Clifford. Consequently, every transversal conversion between the associated $[[2m-1, 1, m]]_q$ quantum MDS encoders is Clifford factor by factor, and none admits a transversal non-Clifford logical unitary. For odd prime q, generalized Reed–Solomon codes of every even length $2m \leq q+1$ attain exactly the full projective one-qudit Clifford group. The proof shortens C and $C^\perp$ to $m+1$ coordinates; the resulting marginal is diagonal on the full local Weyl basis, and its rank-one contractions recover the Weyl axes.

For $m = 3$, a degree-eight quotient z classifies projective, monomial-code, local-Clifford, and local-unitary equivalence in a pencil of six-point arcs. With any party taken as the input of a $[[5, 1, 3]]_q$ encoder, the fixed-party symplectic kernel is $\mathrm{SL}_2(q)$ on the conic/GRS locus and the split torus off it. Operator-valued marginals recover the local Weyl frame and force rigidity, whereas every fixed-copy scalar contraction is generically constant; exact marginal and four-copy witnesses still detect special strata.

1 Introduction

An absolutely maximally entangled state has maximally mixed reductions on every subsystem no larger than half the parties. A linear $[2m, m, m+1]_q$ MDS code supplies such a state at minimum support: if $C \leq \mathbb{F}_q^{2m}$, then

$$|\Psi_C\rangle = q^{-m/2} \sum_{c\in C}|c\rangle \tag{1.1}$$

is AME$(2m, q)$; conversely, the equal-phase state of a linear $[2m, m]_q$ code is AME only if the code is MDS. This code–state dictionary is standard [RGRA18]. Our question is how much of the code geometry survives arbitrary single-party changes of basis.

The local-unitary versus local-Clifford problem is false for general stabilizer states [JCWY10]. It is therefore important that the following result concerns a specific MDS/CSS family and says more than equality of two orbit partitions.

Theorem 1.1 (Local-unitary rigidity)**.** Let q be a prime power, let $m \geq 2$, and let $C, D \leq \mathbb{F}_q^{2m}$ be linear $[2m, m, m+1]_q$ MDS codes. Suppose that, for a party permutation π, single-qudit unitaries $U_1, \ldots, U_{2m}$, and a phase θ,

$$(U_1 \otimes \cdots \otimes U_{2m})\pi|\Psi_C\rangle = e^{i\theta}|\Psi_D\rangle.$$

Then every U_i normalizes the finite-field Weyl system. In particular, the displayed equivalence is local Clifford.

Corollary 1.2 (Transversal Clifford no-go)**.** Let $V_C, V_D : \mathbb{C}^q \to (\mathbb{C}^q)^{\otimes(2m-1)}$ be encoding isometries whose normalized Choi states are $|\Psi_C\rangle, |\Psi_D\rangle$ as in Theorem 1.1. They encode the associated $[[2m-1, 1, m]]_q$ quantum MDS codes. If

$$(U_1 \otimes \cdots \otimes U_{2m-1})V_C = V_D L$$

for a logical unitary L, then L and every U_i are Clifford. Thus every transversal conversion between two codes in the family is Clifford factor by factor. Taking $C = D$, no such code admits a transversal non-Clifford logical unitary.

The mechanism is visible in one reduced operator. Retain any $m+1$ parties and shorten both C and $C^\perp$ on the other $m-1$. Each shortening is $[m+1, 1, m+1]_q$, and together they give a q^2-element stabilizer subgroup. At every retained party its $q^2 - 1$ nonidentity elements run bijectively through the nonidentity Weyl operators. The entire reduced state, including its identity term, is a diagonal $(m+1)$-way tensor on the full q^2-element Weyl basis. A contraction of this tensor has rank one exactly on a coordinate axis. Those axes are intrinsic, so any product unitary carrying one reduced state to another must permute the local Weyl axes. Since conjugation fixes the identity axis, this is precisely the Clifford condition. The corollary follows by applying the theorem to the Choi state of a transversal logical action.

Corollary 1.3 (Diagonal-isodual transversal logical group)**.** Let q be an odd prime, let $m \geq 2$, and let $C \leq \mathbb{F}_q^{2m}$ be a linear $[2m, m, m+1]_q$ MDS code. The projective group of logical unitaries of the associated $[[2m-1, 1, m]]_q$ quantum MDS code that admit tensor-product physical implementations is

$$\begin{cases} \mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q), & \text{if } SC = C^\perp \text{ for a nonsingular diagonal } S, \\ \mathbb{F}_q^2 \rtimes T, & \text{otherwise,} \end{cases}$$

where $T = \{\mathrm{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}$.

The phrase "modulo phase" is essential. Over an odd field the $\mathrm{SL}_2(q)$ factor on the diagonally isodual branch has coherent Weil lifts, so no metaplectic obstruction remains there. The full affine group does not lift through the one-qudit scalar quotient: its translation subgroup is the projective Weyl group, whose two coordinate generators have a nontrivial scalar commutator. This Heisenberg obstruction is independent of the factor set for party-moving symmetries in Corollary 3.5. Generalized and extended generalized Reed–Solomon codes are diagonally isodual, but the condition is not confined to that family: self-dual non-GRS MDS constructions are known [ZL24].

For example, an extended $[8, 4, 5]_7$ Reed–Solomon code gives an $\mathrm{AME}(8, 7)$ tensor and a $[[7, 1, 4]]_7$ quantum MDS code whose projective transversal logical group has order $7^2|\mathrm{SL}_2(7)| = 16464$. No non-Clifford transversal gate enlarges this group.

Rains used the corresponding three-Pauli tensor to determine the automorphisms of qubit distance-two codes [Rai99, Theorem 13], and Van den Nest, Dehaene, and De Moor used that rigidity in their minimal-support criterion for qubit stabilizer states [VdNDDM05]. Our argument is a finite-field extension of this axis-recovery mechanism. The contribution here is its uniform realization for equal-phase CSS states from linear $[2m, m, m+1]_q$ MDS codes, together with the transversal corollary. We do not claim a new LU–LC mechanism in the abstract. The qualifier is essential: arbitrary phased minimal-support $\mathrm{AME}(6, d)$ states form infinitely many LU classes [RL25], and neither nonlinear orthogonal arrays nor general stabilizer states are covered here.

The geometric applications in the remainder of the paper specialize to $m = 3$. Our main application is the ordered pencil

$$H_t = \big((0, 1, 1-t), (0, 1, t-1), (1, 1-t, 0), (1, t-1, 0), (1, 0, -t), (1, 0, t)\big). \tag{1.2}$$

On its admitted odd-prime-field non-GRS locus, define

$$\begin{aligned} A(t) &= -4t(t-1)^2, \\ B(t) &= (t^2-t+1)(t^2-3t+1), \\ z(t) &= \bigl(B(t)/A(t)\bigr)^2. \end{aligned} \tag{1.3}$$

Exact bracket identities show that z is the degree-eight projective quotient of the pencil. Theorem 1.1 upgrades its local-Clifford classification to the following complete local result.

Corollary 1.4 (Classification of the admitted pencil)**.** For admitted parameters t, u over a prime field $\mathbb{F}_q$ with q odd, the following are equivalent:

$$H_t \sim_{\mathrm{proj}} H_u \iff \ker H_t \sim_{\mathrm{mon}} \ker H_u \iff \Psi_t \sim_{\mathrm{LC}} \Psi_u \iff \Psi_t \sim_{\mathrm{LU}} \Psi_u \iff z(t) = z(u),$$

where party permutations are allowed.

The second main result gives this geometry a direct coding-theoretic interpretation over odd prime fields. Regard any one tensor leg as the input of the associated $[[5,1,3]]_q$ encoder. The fixed-party product-Clifford symmetries induce the full logical $\mathrm{SL}_2(q)$ precisely when the six-arc is conic/GRS; otherwise they induce only the split torus. Thus the nineteenth-century self-association condition for six points controls which one-qudit logical Clifford gates are available without moving parties. Including logical Paulis, the exact fixed-party projective logical group is $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ on the GRS locus and $\mathbb{F}_q^2 \rtimes T$ off it.

Quantum Reed–Solomon codes and their encoding circuits were developed by Grassl, Geiselmann, and Beth [GGB99]. Aharonov and Ben-Or constructed fault-tolerant coordinatewise gates for polynomial codes, with degree reduction when a gate changes the polynomial degree [ABO08, Section 5]; Gottesman constructed the full logical Clifford group for any prime-dimensional stabilizer code using ancillas and Pauli measurements [Got99]. These results establish that GRS and polynomial encoders support fault-tolerant logical gates, but they do not identify the fixed-party product-Clifford kernel. Corollary 1.3 combines diagonal code–dual equivalence with LU rigidity to identify the exact transversal logical group for every half-dimensional MDS code. In coding theory, diagonal code–dual equivalence is the fixed-coordinate form of isometry duality; it occurs in algebraic-geometry MDS families as well as GRS codes [ZZ26]. For binary self-dual CSS codes, Tansuwannont, Takada, and Fujii give criteria for transversal Hadamard and phase operations relative to a compatible logical basis [TTF25]. Their setting and conclusion differ from the odd-prime, one-logical-qudit exact fixed-party group here. Theorem 5.1 identifies the geometric condition among six-arcs: self-association is exactly the condition for the kernel to exceed the split torus. Our literature search for this comparison covered isometry-dual MDS constructions, quantum Reed–Solomon and polynomial-code gate constructions, self-dual CSS transversal Clifford gates, general prime-dimensional stabilizer Clifford synthesis, and restrictions on transversal gate sets. We claim the exact group statements, not priority for the underlying code or gate constructions. Eastin and Knill's restriction on universal transversal sets gives the general boundary [EK09].

Classically, we use Coble's association theory [Cob22] through the modern treatment of Dolgachev and Ortland [DO88, Chapter III]; for finite-projective arc background see Segre, Hirschfeld, and the modern survey of Ball and Lavrauw [Seg55, Hir98, BL19].

The rest of the paper separates classification from explicit scalar certificates. Section 2 fixes the arc, code, Weyl, and AME conventions. Section 3 proves the headline theorem and its transversal and GRS corollaries. Section 4 proves the z-classification, and Section 5 proves the logical-Clifford phase theorem: $\mathrm{SL}_2(q)$ on the GRS locus and a split torus off it. Section 6 gives a uniform H3/GRS marginal separator, an exact four-copy $q = 13$ witness, and the generic-constancy obstruction for fixed-copy scalars. Appendix A calculates the exceptional divisor of that witness, redundant given Corollary 1.4; it is not part of the classification. Section 7 records the computational trust boundary and the precise scope of all results.

2 Six-arcs, MDS codes, and CSS states

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x + a\rangle, \qquad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. For the CSS stabilizer we use the exact lift

$$W_{(c,h)} = X(c)Z(h) = \bigotimes_{i=1}^{6} X(c_i)Z(h_i), \qquad (c, h) \in C \oplus C^\perp.$$

Since $h \cdot c' = 0$ for $h \in C^\perp$ and $c' \in C$, these lifts multiply without a residual phase and fix $|\Psi_C\rangle$. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

A six-arc is an ordered sextuple of points of $\mathbb{P}^2(\mathbb{F}_q)$ with no three collinear. Choose nonzero column representatives and collect them in a 3-by-6 matrix H. The arc condition says that every three-column minor of H is nonzero. Consequently

$$C = \ker H \leq \mathbb{F}_q^6$$

is an $[6, 3, 4]_q$ MDS code, and its dual $C^\perp = \operatorname{rowspace} H$ is MDS with the same parameters. Conversely, a parity-check matrix of such a code supplies a six-arc. Projectivities, independent rescaling of columns, and permutation of columns correspond exactly to monomial equivalences of the kernels. These objects exist exactly when $q \geq 4$. Indeed, from one point of a six-arc the five joining lines are distinct, so $q + 1 \geq 5$. For $q = 4$, a conic together with its nucleus is a six-arc; for $q \geq 5$, any six points of a nonsingular conic form a six-arc.

Proposition 2.1 (Arc–AME dictionary)**.** For a six-arc with kernel C, the state $|\Psi_C\rangle$ in (1.1) is a minimal-support stabilizer AME$(6, q)$ state. Its projective Pauli-label Lagrangian is

$$L_C = (C \oplus 0) + (0 \oplus C^\perp) \leq \mathbb{F}_q^{12}.$$

Monomial code equivalence, together with a party permutation, induces local-Clifford equivalence of the corresponding states.

Proof. The operators $X(c)$, $c \in C$, fix the equal superposition. The operators $Z(h)$, $h \in C^\perp$, fix it because $\chi(h \cdot c) = 1$. The two subgroups commute, and their label space has dimension six and is Lagrangian.

For any three-coordinate set I, puncturing C to I is bijective: injectivity follows from distance four and both spaces have dimension three. In the partial trace of $|\Psi_C\rangle\langle\Psi_C|$, equality on the complementary three coordinates therefore forces equality of the codewords, while the projection to I runs once through $\mathbb{F}_q^3$. Hence every three-party reduction is $q^{-3}I$. Smaller reductions are maximally mixed by tracing again, so the state is AME$(6, q)$. Its support has the minimum possible size q^3.

A column multiplier $x \mapsto ax$ is a single-party Clifford operation: it sends $X(b)$ to $X(ab)$ and $Z(b)$ to $Z(a^{-1}b)$. Projective row operations do not change the kernel, and column permutations permute parties. This proves the last assertion. $\square$

For a party set S, let

$$L_C(S) = \{\ell \in L_C : \operatorname{supp}(\ell) \subseteq S\}.$$

The stabilizer partial-trace formula gives

$$\rho_S^C = q^{-|S|} \sum_{\ell \in L_C(S)} W_\ell. \tag{2.1}$$

This formula is the bridge used in both the rigidity proof and the marginal invariants below. A Pauli-labelled quantity is not by itself an LU invariant; the invariant statements will always be expressed either as covariance of the reduced operators or as basis-free tensor contractions.

The GRS boundary has a simple geometric meaning. A dimension-three length-six generalized Reed–Solomon code is represented, up to column multipliers, by six points on a nonsingular conic in $\mathbb{P}^2$. We use "GRS state" for the state of any such code. The non-GRS pencil in Section 4 consists of arc matrices H_t away from that conic locus and from the other factors needed to keep the columns distinct and the displayed coordinates defined.

3 Shortened Weyl tensors and LU rigidity

We first isolate the linear-algebra statement that recovers a diagonal tensor's coordinate axes. It is the full-Weyl counterpart of the three-Pauli argument in [Rai99, VdNDDM05].

Lemma 3.1 (Diagonal-tensor axes)**.** Let $r \geq 3$, let $E_1, \dots, E_r$ be N-dimensional complex inner product spaces, and choose orthonormal bases $\{e_{ij} : 1 \leq j \leq N\}$. If

$$T = \sum_{j=1}^{N} \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \qquad \lambda_j \neq 0, \tag{3.1}$$

then the coordinate axes in every E_i are intrinsic to T. Consequently, a product of invertible linear maps carrying one tensor of the form (3.1) to another is monomial in the displayed bases.

Proof. Contract the first factor against $x = \sum_j x_j e_{1j}^*$. The result is

$$T_x = \sum_j \lambda_j x_j e_{2j} \otimes \cdots \otimes e_{rj}.$$

Flattening from E_2 to $E_3 \otimes \cdots \otimes E_r$ gives a matrix of rank $\#\{j : x_j \neq 0\}$. Thus T_x has rank one exactly when x lies on a coordinate axis. Product equivalence preserves this rank-one contraction locus, and hence permutes the axes in the first factor. Apply the same argument in every factor. ☐

Call an operator on $r \geq 3$ qudits *full-Weyl diagonal* if it has an expansion

$$R = \sum_{v \in \mathbb{F}_q^2} \lambda_v \bigotimes_{i=1}^{r} W_{L_i v}, \qquad L_i \in \mathrm{GL}_2(q), \quad \lambda_v \neq 0. \tag{3.2}$$

The coefficients need not be equal. What matters is that the same q^2-element index set runs bijectively through the complete local Weyl basis at every party.

Proposition 3.2 (Full-Weyl marginal criterion)**.** Let R and R' be full-Weyl diagonal operators on the same $r \geq 3$ parties, possibly with different maps L_i and coefficients. If

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger,$$

then every U_i is Clifford.

Proof. View R and R' as tensors in the local Hilbert–Schmidt spaces. Both have the form in Lemma 3.1, with $N = q^2$. Local conjugation acts invertibly on each factor, so it must permute the Weyl axes. It fixes the identity operator and hence the identity axis. It therefore permutes the nonidentity projective Weyl operators, which is exactly the single-qudit Clifford condition. □

Corollary 3.3 (Full-Weyl marginal-cover rigidity)**.** Let $|\psi\rangle, |\phi\rangle$ be n-qudit states and suppose a product unitary, after a party relabelling, carries $|\psi\rangle$ to $|\phi\rangle$ up to phase. If every party belongs to a set S of at least three parties for which both reduced operators ρ_S^ψ and ρ_S^ϕ are full-Weyl diagonal, then every local factor is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each chosen S. Apply Proposition 3.2; the chosen sets cover all parties. □

Let $C \leq \mathbb{F}_q^{2m}$ be $[2m, m, m+1]_q$ MDS, and fix a party set S of size $m+1$. Shortening C on the $m-1$ coordinates outside S gives an $[m+1, 1, m+1]_q$ MDS code. The dual $C^\perp$ has the same parameters, so its corresponding shortening does as well. Choose nonzero shortened words $c^S \in C$ and $h^S \in C^\perp$. Each has support exactly S, and

$$L_C(S) = \{(ac^S, bh^S) : a, b \in \mathbb{F}_q\}. \tag{3.3}$$

At a party $i \in S$, projection of this plane to the local Pauli labels is

$$(a, b) \longmapsto (ac_i^S, bh_i^S),$$

which is invertible because both coordinates are nonzero. Thus all q^2 labels in $L_C(S)$ run bijectively through the full local Weyl basis at every party of S; in particular, the $q^2 - 1$ nonidentity labels run bijectively through the nonidentity Weyl labels.

Proposition 3.4 (A shortened marginal fixes the local Weyl axes)**.** Let $C, D \leq \mathbb{F}_q^{2m}$ be linear $[2m, m, m+1]_q$ MDS codes. If a product unitary on an $(m+1)$-set S carries ρ_S^C to ρ_S^D, then every local factor on S is Clifford.

Proof. For a general equal-phase CSS state, the same stabilizer partial-trace calculation as in (2.1) gives

$$\rho_S^C = q^{-|S|} \sum_{\ell \in L_C(S)} W_\ell.$$

View the entire marginal as a tensor in the $m+1$ local Hilbert–Schmidt spaces, each with the full q^2-element Weyl basis. By (3.3), after indexing $v = (a, b)$, it has the form

$$\rho_S^C = \sum_{v \in \mathbb{F}_q^2} \lambda_v \bigotimes_{i \in S} W_{L_i v}, \tag{3.4}$$

where every L_i is invertible and every coefficient λ_v is nonzero. (With the normalization above, all coefficients are $q^{-|S|}$.) This is a diagonal tensor with $N = q^2$, including the identity label $v = 0$.

The same calculation for D makes ρ_S^D full-Weyl diagonal. The conclusion is Proposition 3.2. □

Proof of Theorem 1.1. Absorb the party permutation by relabelling D. Partial trace of the $2m$-party equivalence gives a product-unitary equivalence $\rho_S^C \sim \rho_S^D$ for every $(m+1)$-set S. Given a party i, choose such a set containing it and apply Proposition 3.4. Thus every U_i is Clifford. □

Proof of Corollary 1.2. Let $|\Phi\rangle = q^{-1/2}\sum_x |x,x\rangle$, and use the Choi convention $|\Psi_C\rangle = (I \otimes V_C)|\Phi\rangle$ and $|\Psi_D\rangle = (I \otimes V_D)|\Phi\rangle$. Writing $U_{\text{phys}} = \bigotimes_i U_i$, the relation $U_{\text{phys}} V_C = V_D L$ gives

$$(I \otimes U_{\text{phys}})|\Psi_C\rangle = (L^T \otimes I)|\Psi_D\rangle.$$

Hence

$$\big((L^T)^{-1} \otimes U_{\text{phys}}\big)|\Psi_C\rangle = |\Psi_D\rangle.$$

Theorem 1.1 makes every displayed local factor Clifford. Entrywise conjugation preserves the finite-field Weyl group, since $\overline{X(a)} = X(a)$ and $\overline{Z(b)} = Z(-b)$; it therefore preserves the Clifford normalizer. Taking adjoints preserves the normalizer as well, and $U^T = (\overline{U})^\dagger$. Thus transposition preserves the finite-field Clifford group, and L is Clifford. $\square$

Corollary 3.5 (Discrete product-unitary symmetry)**.** For every equal-phase MDS–CSS state in Theorem 1.1, the product-unitary automorphism group modulo one-site scalar phases is finite and discrete. Its identity component before that quotient consists only of one-site scalar phases. More precisely, if $T = (S^1)^{2m}$, G is the fixed-party product-unitary automorphism group, and $\widetilde{G}$ also displays party permutations, then

$$1 \longrightarrow T \longrightarrow G \longrightarrow \Gamma \longrightarrow 1, \qquad 1 \longrightarrow T \longrightarrow \widetilde{G} \longrightarrow \widetilde{\Gamma} \longrightarrow 1$$

are short exact sequences with closed connected kernel and finite discrete quotient; both inclusions of T are continuous. In particular, G and $\widetilde{G}$ are finite unions of connected components, each a translate of T. If Π is the subgroup of party permutations realized by $\widetilde{G}$, there is a further exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \widetilde{\Gamma} \longrightarrow \Pi \longrightarrow 1.$$

This last extension splits exactly when the realized permutations admit a homomorphic choice of projective product-unitary lifts. It has a section-free outer action $\Pi \to \mathrm{Out}(\Gamma)$: conjugation by any lift of $\pi \in \Pi$ gives an automorphism of Γ, and changing the lift changes that automorphism by an inner automorphism. For a normalized set-theoretic section $s(1) = 1$, write

$$s(\pi)s(\rho) = f_s(\pi,\rho)s(\pi\rho), \qquad f_s(\pi,\rho) \in \Gamma.$$

Then $f_s(1,\pi) = f_s(\pi,1) = 1$, and, with α_π denoting conjugation by $s(\pi)$,

$$f_s(\pi,\rho)f_s(\pi\rho,\upsilon) = \alpha_\pi(f_s(\rho,\upsilon))\, f_s(\pi,\rho\upsilon).$$

If another normalized section is $s'(\pi) = b(\pi)s(\pi)$, where $b(1) = 1$, then

$$f_{s'}(\pi,\rho) = b(\pi)\alpha_\pi(b(\rho))\, f_s(\pi,\rho)b(\pi\rho)^{-1}.$$

The order of these factors is essential because Γ need not be abelian. The extension splits exactly when this factor set can be changed to the constant identity factor set by such a normalized b.

Proof. Every local factor of an automorphism is Clifford. The projective single-qudit Clifford group over a finite field is finite, as is the party-permutation group. The kernel of projectivizing each local factor is exactly the torus of one-site scalar phases. That torus is the identity component, hence is closed; the finite Hausdorff quotient is discrete. Translation identifies every connected component with the scalar torus, so finitely many quotient representatives cover the group. Forgetting the local factors maps $\widetilde{\Gamma}$ onto the realized permutation subgroup Π. Its kernel consists exactly of the fixed-party projective automorphisms, which gives the third sequence. A splitting is, by definition, a group-homomorphic right inverse; arbitrary or generator-by-generator lifts do not provide one. Conjugation by the middle group descends to the outer action because elements of the kernel act by inner automorphisms. Associativity of three chosen lifts gives the displayed factor-set identity. Substituting $s'(\pi) = b(\pi)s(\pi)$ gives the change-of-section formula without commuting any kernel factors. Finally, the factor set is identically one precisely when the section preserves multiplication, which proves the splitting characterization. $\square$

Proof of Corollary 1.3. Write a fixed-party product-Clifford action on Pauli labels as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix},$$

and let $D_\alpha, D_\beta, D_\gamma, D_\delta$ be the diagonal matrices with the corresponding entries. Preservation of $L_C = C_X \oplus C_Z^\perp$ is equivalent to

$$\begin{aligned} D_\alpha C &\subseteq C, \quad & D_\gamma C &\subseteq C^\perp, \\ D_\beta C^\perp &\subseteq C, \quad & D_\delta C^\perp &\subseteq C^\perp. \end{aligned} \tag{3.5}$$

We first isolate the only use of the MDS hypothesis. If E, F are $[2m, m, m+1]_q$ MDS codes and a diagonal matrix D satisfies $DE \subseteq F$, then either $D = 0$, or D is nonsingular and $DE = F$. Indeed, if the support of a nonzero D had size at most m, choose a word of E nonzero at a supported coordinate; its image would be a nonzero word of F of weight at most m. Thus D has at least $m+1$ nonzero entries. Its kernel on E consists of words supported on at most $m-1$ coordinates, so it is zero. Equality follows from the dimensions. A zero diagonal entry would then force every word of F to vanish in that coordinate, contrary to the MDS property.

If the block at the chosen input is nondiagonal, then D_β or D_γ is nonzero. The lemma and (3.5) give

$$D_\gamma C = C^\perp \qquad \text{or} \qquad D_\beta C^\perp = C.$$

The second equality can be inverted, so either one supplies a nonsingular diagonal S with $SC = C^\perp$. Consequently, on the non-isodual branch every input block is diagonal. Its determinant is one, and the common blocks $\mathrm{diag}(a, a^{-1})$ always preserve L_C; hence the exact linear factor there is T.

Now suppose

$$S = \mathrm{diag}(s_1, \ldots, s_{2m}), \qquad SC = C^\perp. \tag{3.6}$$

For $\lambda, \mu \in \mathbb{F}_q$, put

$$F_i^-(\lambda) = \begin{pmatrix} 1 & 0 \\ \lambda s_i & 1 \end{pmatrix}, \qquad F_i^+(\mu) = \begin{pmatrix} 1 & \mu s_i^{-1} \\ 0 & 1 \end{pmatrix}. \tag{3.7}$$

On the CSS label space $L_C = (C \oplus 0) + (0 \oplus C^\perp)$, the product of the lower blocks sends

$$(c, h) \longmapsto (c, h + \lambda S c),$$

and the product of the upper blocks sends

$$(c, h) \longmapsto (c + \mu S^{-1} h, h).$$

Thus both products preserve L_C and lift to product-Clifford tensor symmetries after product-Pauli corrections of their stabilizer phases. At any chosen input party j, their blocks run through all lower and upper elementary unipotents because $s_j \neq 0$. Together these generate $\mathrm{SL}_2(q)$. Under the Choi correspondence, an input block A induces the logical block A^{-T}. This map is an automorphism of $\mathrm{SL}_2(q)$, so every one-qudit logical symplectic action has a product-Clifford physical implementation. Logical Paulis have physical Pauli representatives, so the full projective one-qudit Clifford group $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ is transversal.

Corollary 1.2 excludes every product-unitary logical action outside that group. Hence the displayed group is exact.

For a generalized or extended generalized Reed–Solomon code, the standard dual formula supplies (3.6), including the extended evaluation point at length $q+1$. Thus the GRS tower lies on the diagonally isodual branch, but the proof used no evaluation structure.

There are two different lift questions behind this projective statement. Fix an input coordinate j, put $t_i = s_i/s_j$, and write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(q).$$

Propagation to coordinate i is

$$\phi_i(A) = \begin{pmatrix} a & t_i^{-1}b \\ t_i c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & t_i \end{pmatrix} A \begin{pmatrix} 1 & 0 \\ 0 & t_i^{-1} \end{pmatrix}. \tag{3.8}$$

Thus every ϕ_i is a homomorphism, ϕ_j is the identity, and the elementary matrices in (3.7) satisfy all $\mathrm{SL}_2(q)$ relations coordinatewise. This calculation also applies at the extended evaluation point. It uses only the diagonal duality $SC = C^{\perp}$, not the evaluation formula for a GRS code.

For odd q, the finite Stone–von Neumann representation of the Heisenberg group H_q extends to a genuine representation of $\mathrm{SL}_2(q) \ltimes H_q$ [Gér76, Theorem 1]. Apply its Weil factor to each $\phi_i(A)$. The resulting tensor product is multiplicative in A, and exact Egorov covariance preserves the CSS stabilizer line. Consequently the one-qudit Clifford scalar extension, restricted to the linear $\mathrm{SL}_2(q)$ factor, splits coherently with the propagated tensor symmetries. The exceptional uniqueness clause at $q = 3$ in Gérardin's theorem does not remove existence.

The affine extension does not split. Choose $a, b \in \mathbb{F}_q$ with $\chi(ab) \neq 1$, and put $X = X(a)$, $Z = Z(b)$. Their projective classes commute, whereas every choice of unitary representatives satisfies

$$ZX = \chi(ab)XZ, \qquad \chi(ab) \neq 1.$$

Scalar changes of X and Z do not change their commutator. Hence the translation subgroup $\mathbb{F}_q^2$ has no homomorphic lift to the one-qudit unitary Clifford group. The coherent odd-field lift is therefore $H_q \rtimes \mathrm{SL}_2(q)$, not a section of $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ through the scalar quotient. This is the Heisenberg central extension, not a residual metaplectic or Schur-multiplier obstruction on $\mathrm{SL}_2(q)$.

Neither conclusion decides the party-permutation extension in Corollary 3.5. Its quotient is the evaluation-set-dependent realized permutation group Π, and its kernel is the full fixed-party projective group Γ, rather than the scalar phases. Its splitting remains exactly the nonabelian factor-set triviality criterion stated there. □

For a stabilizer state, the hypotheses of Corollary 3.3 hold whenever every party belongs to a support S of size at least three whose full supported stabilizer label subgroup is an $\mathbb{F}_q$-linear plane and projects isomorphically onto the local Weyl-label plane at every party of S.

4 The non-GRS pencil and its quotient

Let H_t be the ordered matrix in (1.2), over a finite field of odd order, and put

$$G(t) = t^4 - 4t^3 + 7t^2 - 4t + 1.$$

The admitted non-GRS condition is

$$t(t-1)(t^2-t+1)(t^2-3t+1)G(t) \neq 0. \tag{4.1}$$

Factoring the twenty 3-by-3 column minors gives the first four factors, which keep the arc and quotient coordinates nondegenerate. The determinant for a conic through the six columns is a unit times G, so $G = 0$ is exactly the GRS boundary. Write $C_t = \ker H_t$ and $\Psi_t = \Psi_{C_t}$, and let A, B, z be as in (1.3).

Theorem 4.1 (Projective and Clifford quotient)**.** For admitted t, u, the following are equivalent:

$$H_t \sim_{\text{proj}} H_u, \qquad C_t \sim_{\text{mon}} C_u, \qquad z(t) = z(u),$$

allowing arbitrary party permutations. If q is prime, these are also equivalent to $\Psi_t \sim_{\text{LC}} \Psi_u$.

Proof. Set

$$y(t) = \frac{(t-1)^2}{t}.$$

Then $A = -4t^2 y$, $B = t^2(y^2-1)$, and

$$z = \frac{(y-y^{-1})^2}{16}.$$

It follows that $z(t) = z(u)$ exactly when

$$y(u) \in \{y(t), -y(t), y(t)^{-1}, -y(t)^{-1}\}. \tag{4.2}$$

Each of the four relations is realized by a direct projectivity and a party permutation. In order, one may use

relation	projectivity
$y(u) = y(t)$	$\operatorname{diag}(1, (1-u)/(1-t), u/t)$
$y(u) = -y(t)$	$\operatorname{diag}(1, (1-u)/(1-t), -u/t)$
$y(u) = y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (u-1)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$
$y(u) = -y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (1-u)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}.$

The corresponding permutations are respectively the identity, $(5\,6)$, $(3\,5)(4\,6)$, and $(3\,5\,4\,6)$. All denominators and determinants are nonzero on the admitted locus.

Conversely, the ten signed products of complementary 3-by-3 brackets of H_t form the multiset

$$\{+A, +A, +A, -A, -A, -A, +B, +B, -B, -B\}. \tag{4.3}$$

Projectivity and independent column rescaling multiply all ten products by one common scalar; party permutation changes at most their common sign. On the admitted locus $A, B \neq 0$. If $A^2 \neq B^2$, the multiplicities three and two in (4.3) distinguish the two opposite pairs and recover $(B/A)^2 = z$. If $A^2 = B^2$, the multiset instead collapses to two opposite values, each of multiplicity five. That collapsed pattern is itself invariant under the common scaling and sign, and it is equivalent to $z = 1$. Thus the multiset recovers z in both cases, and projective equivalence forces equality of z. The equivalence with monomial codes follows from Section 2, and projective equivalence supplies an LC equivalence.

For the remaining direction assume that q is prime. Then a local Clifford induces a block $F_i \in \mathrm{SL}_2(q)$ on each local Pauli-label plane. Put $K_T = L_{C_t}(T)$ for every four-set T. If $i, j \in T$, projection $p_i : K_T \to \mathbb{F}_q^2$ onto the Pauli-label plane at i is an isomorphism by (3.3). Define the transition

$$M_T(i,j) = p_j p_i^{-1} : (\mathbb{F}_q^2)_i \longrightarrow (\mathbb{F}_q^2)_j$$

and, for distinct four-sets T, U containing i, j, the holonomy

$$\mathrm{Hol}(T,U;i,j) = M_U(j,i) M_T(i,j). \tag{4.4}$$

Local symplectic blocks F_i send this endomorphism to $F_i \,\mathrm{Hol}\, F_i^{-1}$. Reversing T, U takes its inverse, and a party permutation only relabels the data. Thus the multiset of

$$r(M) = \mathrm{Tr}(M)^2 / \det M$$

over the $15 \cdot 6 \cdot 5 = 450$ choices is an LC and party-permutation invariant.

Direct substitution of the shortened generators into (4.4) gives the following rational-function multiset; the symbolic identities and a direct-Lagrangian reconstruction are independently replayed by the paper-local certificate described in Section 7:

$$\begin{array}{c|l} 4 & 90 \\ -1/z & 144 \\ (2z+1)^2/z^2 & 24 \\ \text{each root of } X^2 - 8X + (8 - 16z - 1/z) & 96. \end{array} \tag{4.5}$$

Here 96 is the multiplicity of each root before specialization; coincident values have their multiplicities added. The identity

$$4B(t)^2 + A(t)^2 = 4G(t)^2$$

shows that the excluded GRS locus is exactly $z = -1/4$. Away from it, the constant 4 bin is isolated and the two quadratic roots are distinct. The bin containing $-1/z$ can merge with neither, one, or both of the weight-24 and one weight-96 contributions, so its multiplicity is $144, 168, 240$, or 264. No other bin has one of these multiplicities. Hence the multiset canonically recovers $-1/z$, including every modular collision, and LC equivalence forces equality of z. ☐

The map $t \mapsto z$ has degree eight: the four transformations in (4.2), followed by the quadratic relation $t + t^{-1} = y + 2$, account for its eight sheets. The signed refinement $w = B/A$, $z = w^2$, is not quantum data. Both even and odd explicit party permutations reverse w, and their projectivities lift to monomial local Cliffords.

Proof of Corollary 1.4. Theorem 4.1 gives all equivalences except LU. Every LC is an LU, while Theorem 1.1 turns every LU intertwiner between the states Ψ_t, Ψ_u into an LC intertwiner. ☐

The prime-field hypothesis is necessary for this scalar classification. Over $\mathbb{F}_{p^e}$, the basis permutation $|x\rangle \mapsto |x^p\rangle$ is a local Clifford and sends Ψ_t to Ψ_{t^p}, while $z(t^p) = z(t)^p$ need not equal $z(t)$. Concretely, in $\mathbb{F}_{25} = \mathbb{F}_5[s]/(s^2 - 2)$, both s and $-s = s^5$ are admitted, but

$$z(s) = 3 + 3s, \qquad z(-s) = 3 + 2s.$$

Thus extension-field LC/LU classification requires at least a Frobenius quotient and, for the full Weyl normalizer, potentially further $\mathbb{F}_5$-symplectic identifications. No such classification is claimed here.

5 Diagonal isoduality and the logical-Clifford phase

Throughout this section q is an odd prime. Regard any one party of an AME$(6, q)$ stabilizer tensor as an input and the other five as outputs. This is a $[[5, 1, 3]]_q$ encoding isometry. In prime local dimension, the symplectic action of a one-qudit Clifford is a block in $\mathrm{SL}_2(q)$, so the kernel of the party action on Clifford tensor symmetries is

$$\mathcal{K}_C = \{(F_1, \ldots, F_6) \in \mathrm{SL}_2(q)^6 : (\bigoplus_i F_i) L_C = L_C\}.$$

For $q = p^e$ with $e > 1$, the full local Clifford group acts instead through $\mathrm{Sp}_{2e}(\mathbb{F}_p)$; the displayed field-linear block group is then not the full local Clifford kernel. For an encoder view, the *fixed-party symplectic kernel* means the image of $\mathcal{K}_C$ on its input block. Corollary 1.3 identifies the all-length phase boundary as diagonal isoduality. For six-arcs, classical self-association turns that intrinsic code condition into the conic criterion below.

Theorem 5.1 (Fixed-party logical phase)**.** For an equal-phase CSS state from a six-arc over an odd prime field, the fixed-party symplectic kernel is

$$\begin{cases} \mathrm{SL}_2(q), & \text{if the arc is conic/GRS,} \\ T = \{\mathrm{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}, & \text{if the arc is nonconic.} \end{cases} \tag{5.1}$$

Thus every encoder view of a GRS tensor realizes the full one-qudit logical symplectic group from fixed-party symmetries. Off the GRS locus, the fixed-party symplectic group is the split torus. Any party-moving isoduality that exchanges the CSS X- and Z-spaces acts by the nontrivial Weyl-group element and generates the normalizer of T together with T.

Proof. Every four-party support carries the two-dimensional shortened label plane (3.3), and projection to the Pauli plane of any retained party is an isomorphism. If F_i is fixed at one party, requiring $\bigoplus_i F_i$ to carry each shortened plane to itself therefore determines the other five blocks through overlapping four-sets. These conditions are also sufficient: the shortened planes span L_C, because the weight-four shortened words span both three-dimensional MDS codes C and $C^\perp$.

The common diagonal anchor $\mathrm{diag}(a, a^{-1})$ preserves $C_X \oplus C_Z^\perp$, so T is always present. Write a propagated collection as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix}, \qquad D_\alpha = \mathrm{diag}(\alpha_1, \ldots, \alpha_6), \quad \text{and similarly for } D_\beta, D_\gamma, D_\delta.$$

Preservation of $L_C = C_X \oplus C_Z^\perp$ is equivalent to

$$\begin{aligned} D_\alpha C \subseteq C, &\quad D_\gamma C \subseteq C^\perp, \\ D_\beta C^\perp \subseteq C, &\quad D_\delta C^\perp \subseteq C^\perp. \end{aligned} \tag{5.2}$$

We use the following elementary MDS fact to control the off-diagonal blocks. If $E, F \leq \mathbb{F}_q^6$ are $[6, 3, 4]$ MDS codes and a diagonal matrix D satisfies $DE \subseteq F$, then either $D = 0$ or D is nonsingular and $DE = F$. Indeed, if $D \neq 0$ and its diagonal support had size at most three, choose $e \in E$ nonzero at a supported coordinate. Then De would be a nonzero word of F of weight at most three. Thus D has at least four nonzero entries. Its kernel on E consists of words supported on at most the two remaining coordinates, so the distance-four property makes this kernel zero. Consequently $DE = F$. If a diagonal entry of D were zero, every word of F would vanish in that coordinate, which is impossible for an MDS code.

Apply this fact to $D_\gamma C \subseteq C^\perp$ and $D_\beta C^\perp \subseteq C$. A nondiagonal input block has a nonzero off-diagonal entry, hence the corresponding diagonal matrix is nonzero, nonsingular, and gives

$$D_\gamma C = C^\perp \qquad \text{or} \qquad D_\beta C^\perp = C. \tag{5.3}$$

Conversely, if a nonsingular diagonal S satisfies $SC = C^\perp$, then $S^{-1}C^\perp = C$. Taking $D_\gamma = S$, respectively $D_\beta = S^{-1}$, with identity diagonal blocks in (5.2) realizes the lower and upper elementary unipotent input anchors. Together with T, these generate $\mathrm{SL}_2(q)$.

It remains to identify when the diagonal equivalence $C \sim C^\perp$ occurs. A parity-check matrix for C and a kernel basis for it represent associated, or Gale-dual, ordered six-arcs. The classical self-association theorem for six points in $\mathbb{P}^2$ says that these two sextuples are projectively equivalent exactly when the six original columns lie on a conic [Cob22][DO88, Chapter III] [HMSV08, Section 2.3]. Equivalently, solving the multipliers in (5.2) leaves the determinant of the six quadratic evaluation vectors. Since the columns form an arc, the conic is nonsingular. On it, the standard GRS dual multipliers realize the code–dual equivalence and hence every anchor in $\mathrm{SL}_2(q)$; off it, the preceding implication excludes every nondiagonal anchor.

Finally, an isoduality interchanges the two summands C_X and $C_Z^\perp$, so its input block swaps the two Weyl axes. It conjugates a to a^{-1} in T, giving $N(T)$ whenever such a party-moving symmetry is present. □

Corollary 5.2 (Exact fixed-party transversal group)**.** For the associated $[[5,1,3]]_q$ encoder, the projective logical Clifford group realized without moving parties is

$$\begin{cases} \mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q), & \text{if the six-arc is conic/GRS,} \\ \mathbb{F}_q^2 \rtimes T, & \text{if the six-arc is nonconic.} \end{cases}$$

Proof. Every logical Pauli has a physical Pauli representative, giving the normal translation subgroup $\mathbb{F}_q^2$. Theorem 5.1 identifies the exact linear factor realized by fixed-party tensor symmetries. Modulo scalar phases, the one-qudit Clifford group is the affine semidirect product $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$, so these two ingredients give the displayed equalities. □

This is an operational distinction, not a classical automorphism count. The fixed-party non-GRS group has order $q^2(q-1)$, while the GRS group has order $q^2|\mathrm{SL}_2(q)|$. At $q = 11$, the two non-GRS pencil classes also have party-moving isodualities, so their party-moving logical group is $q^2N(T)$; the GRS and party-moving non-GRS orders are 159720 and 2420, respectively.

The MDS property also makes minimum operator pushing uniform. For any input party and nonidentity Pauli label, each of the ten four-sets containing that party gives a unique stabilizer representative supported there. Removing the input leaves a weight-three output Pauli. Thus each input Pauli has exactly ten minimum-support pushes, although the symmetry orbits of those pushes can distinguish finer projective classes. This uses only the projection isomorphism for the shortened planes, not the finite group calculation.

Arithmetic symmetry jumps do not change the phase. The characteristic-17 tetrahedral specialization lies on the GRS boundary and therefore retains the full $\mathrm{SL}_2(17)$ kernel. The characteristic-31 A_5 specialization is non-GRS and therefore retains only the split torus. These are symmetry enhancements inside the two phases, not exceptions to (5.1).

6 Explicit local-unitary invariants

Theorem 1.1 settles equivalence without scalar invariants. Scalar witnesses remain useful because they are short, basis-free certificates and because their failures explain what the operator-valued marginal retains.

6.1 A marginal-moment separator

For a four-party set T, let ρ_T be the reduced state and embed it into all six parties as

$$A_T = \rho_T \otimes I_{T^c}.$$

For an unordered triple $\{T, U, V\}$ of distinct four-sets, the number $\mathrm{Tr}(A_T A_U A_V)$ is invariant under arbitrary product unitaries; party permutation merely permutes the 455 triples. By (2.1),

$$\mathrm{Tr}(A_T A_U A_V) = q^{-\operatorname{rank}(L_C(T)+L_C(U)+L_C(V))}. \tag{6.1}$$

Here and below, ranks of label spaces and matching systems are over $\mathbb{F}_q$. Indeed, the three stabilizer sums have total normalization q^{-12}. Only triples of labels with sum zero survive the full trace; isotropy of L_C removes multiplication phases, and the kernel of the addition map from the three two-planes has size q^{6-r}, where r is the $\mathbb{F}_q$-rank in (6.1).

Index a four-set by its omitted pair, an edge of K_6. The rank in (6.1) is four exactly when the three corresponding chord lines concur in both the six-arc $\mathcal{A}$ and its Gale dual $\mathcal{A}^*$, represented by any kernel basis of the arc matrix. To see this, split each shortened CSS plane into its one-dimensional $X(C)$ and $Z(C^\perp)$ parts. For three omitted edges, the three shortened X-lines span dimension two precisely when the corresponding chords of $\mathcal{A}^*$ concur; the Z-lines give the same criterion for $\mathcal{A}$. Each span has dimension at least two, so their sum has rank four exactly when both concurrences hold.

Sixty triples are stars and concur universally. Any other three-edge graph that contains two adjacent edges but is not a star has two chords meeting at their common arc point and a third chord missing it, since the columns form an arc. The only remaining graph type is a perfect matching. Hence

$$\#\{q^{-4} \text{ moments}\} = 60 + b(\mathcal{A}, \mathcal{A}^*), \tag{6.2}$$

where b counts perfect matchings concurrent in both configurations.

Theorem 6.1 (Uniform H3/GRS separation)**.** Let $\tau^2 = \tau + 1$, and let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\tau]$ of odd residue characteristic for which the reduction of the integral H3 (icosahedral) sextuple

$$((0, 1, 1-\tau), (0, 1, \tau-1), (1, 1-\tau, 0), (1, \tau-1, 0), (1, 0, -\tau), (1, 0, \tau))$$

is a six-arc. Write $k = \mathbb{Z}[\tau]/\mathfrak{p}$. If the reduced arc is non-GRS, its equal-phase state is not LU-equivalent, even after party permutation, to any six-point GRS state over k. (The hypothesis is vacuous in characteristic five, where the reduction is itself GRS.)

Proof. Exact integral chord determinants give ten common concurrent matchings for the H3 arc and its Gale dual. The other five determinant norms are -64 and -4, so the count is exactly ten in every residue reduction covered by the theorem. This is an exact fifteen-matching calculation in $\mathbb{Z}[\tau]$, not a field sample. The H3 moment multiset therefore contains 70 copies of q^{-4}.

For six points S on a conic, a concurrent perfect matching is the orbit pairing of a fixed-point-free involution in $\mathrm{Stab}_{\mathrm{PGL}_2(q)}(S)$. A projectivity fixing three points is the identity, so this group acts freely on ordered triples and its order divides 120. Dickson's classical finite-field treatment covers both the p-regular and p-irregular families [Dic01, §§239–261]; Faber gives a clean modern classification over a general field of characteristic p [Fab23]. The resulting bounds are as follows. Cyclic, dihedral, A_4, and S_4 subgroups have at most $1, 6, 3, 6$ fixed-point-free involutions, respectively. In the faithful degree-six action of A_5, every involution fixes two points. In characteristic three, a semi-elementary case has normal 3-group of order

3: its cyclic complement embeds in $\mathrm{Aut}(C_3)$, so the group is C_3 or S_3 and has at most three involutions. The only subfield groups of order dividing 120 are $\mathrm{PSL}_2(3) \cong A_4$ and $\mathrm{PGL}_2(3) \cong S_4$, already covered above. Thus, outside characteristic five, there are at most six fixed-point-free involutions. The bound is attained whenever $\mathrm{PGL}_2(q)$ contains an octahedral S_4, by the corresponding six-set: the six half-turns about axes through pairs of opposite edges act without fixed vertices. Thus a GRS state has at most 66 copies of q^{-4}, and the margin $70 > 66$ is sharp for this argument. In characteristic five no subgroup bound is needed: $X^2 - X - 1 = (X-3)^2$, so the residue of τ is 3, and direct substitution gives $G(\tau) = 0$. The H3 reduction is therefore GRS, contrary to the theorem's non-GRS hypothesis. Equation (6.1) is LU-invariant, so the gap proves the claim. □

The five nonconcurrent H3 matchings form a one-factorization of K_6. The marginal incidence recovers its S_5 stabilizer but forgets the even A_5 orientation. This is necessary: full local Fourier transform sends Ψ_C to $\Psi_{C^\perp}$, and an integral isoduality returns the H3 code through an odd pentad permutation in every odd reduction. The orientation is an LC-forgettable CSS electric/magnetic marking, not a separate LU class.

6.2 The exact dimension-thirteen four-copy witness

For r ket copies and r bra copies, choose a permutation $\sigma_i \in S_r$ at each party and contract ket copy a with bra copy $\sigma_i(a)$. If G generates a three-dimensional code, let $M_{\boldsymbol{\sigma}}(G)$ be the homogeneous matching system

$$(u_a G)_i = (v_{\sigma_i(a)} G)_i \quad (1 \le i \le 6,\ 1 \le a \le r).$$

Counting its solutions gives the normalized contraction

$$I_{\boldsymbol{\sigma}}(\Psi_C) = q^{3r - \operatorname{rank} M_{\boldsymbol{\sigma}}(G)}. \tag{6.3}$$

Theorem 6.2 (Exact four-copy separator at $q = 13$)**.** The two admitted $q = 13$ pencil classes with $z = 4$ and $z = 12$ are not local-unitary equivalent, even after arbitrary party permutation.

Proof. At $q = 13$, the admitted pencil has two classes in the same complete three-copy and marginal-moment bucket, with $z = 4$ and $z = 12$. Use the explicit four-copy pattern

$$\boldsymbol{\sigma} = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

For $\pi \in S_6$, put $(\pi\boldsymbol{\sigma})_i = \sigma_{\pi^{-1}(i)}$, and sum (6.3) over all party permutations:

$$J_{\boldsymbol{\sigma}}(\Psi) = \sum_{\pi \in S_6} I_{\pi\boldsymbol{\sigma}}(\Psi).$$

Over $\mathbb{F}_{13}$, the $z = 4$ class has rank 21 for all 720 terms. The $z = 12$ class has rank 20 for 192 terms and rank 21 for 528. Therefore

$$J_{\boldsymbol{\sigma}}(z=4) = 720 \cdot 13^{-9}, \qquad J_{\boldsymbol{\sigma}}(z=12) = 3024 \cdot 13^{-9}. \tag{6.4}$$

This proves arbitrary-LU inequivalence directly. The complete two- and three-copy contraction sets do not separate this pair; (6.4) makes no claim that four copies are globally minimal. □

6.3 Why fixed-copy scalars cannot recover the pencil coordinate

Proposition 6.3 (Generic constancy)**.** Fix r and a field k of cardinality q, which is also the local Hilbert-space dimension under the code–state dictionary. Fix an irreducible reduced parameter variety X over k, and a regular generator-matrix chart $G(x)$ of constant code dimension. Every degree-(r,r) permutation contraction of the associated equal-phase states is constant on a dense open subset of X. The same holds for party-orbit sums and linear combinations. In the stable range $q \geq r$, every scalar LU-invariant of bidegree (r,r) is generically constant. For a reducible family, the statement holds separately on each irreducible component.

Proof. Formula (6.3) depends only on the rank of a matrix whose entries are regular functions of the family parameter. A maximal nonzero minor shows that this rank is constant on a dense open set. There are only finitely many diagrams at fixed r, so their generic-rank opens have a common dense intersection because X is irreducible. Over the complex state space, the first fundamental theorem for unitary invariants says that local copy-permutation diagrams span the bidegree-(r,r) invariants; when the local Hilbert-space dimension q is at least r, there are no additional dimension relations among these diagrams. This proves the last statement. □

The dense open here is geometric. It need not contain a rational point over a particular small finite field; the assertion about such points is obtained only after checking that the open has an $\mathbb{F}_q$-point.

For the pencil viewed over its rational-function field, the generic four-copy contraction values therefore generate only the constant field, not $\mathbb{Q}(z)$. The witness (6.4) detects a rank-jump stratum. In contrast, the four-party operator tensor (3.4) retains the Weyl axes and forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

7 Verification, literature, and scope

The rigidity theorem, the arc–AME dictionary, and generic constancy are conceptual proofs in the text. The projective quotient uses displayed bracket identities and projectivities. Its transition-map invariance and collision argument are proved in Section 4, while the 450 exact rational functions are certificate-supported. Section 5 proves the MDS diagonal-multiplier and Gale-duality implications; its $q = 11, 17, 31$ group evaluations are certificate-supported examples, not inputs to the uniform theorem. Section 6 proves the marginal trace/rank and incidence bridges; the H3 determinants and $q = 13$ contractions are certificate-supported. Appendix A proves the systematic transport reduction; the cycle-cover determinants, assignment exhaustion, exact minors, and double-coset support sets are certificate-supported.

The reproducibility package is in `supplement/`. From that directory,

`python3 verify.py`	checks all fifteen byte counts and SHA-256 digests,
`python3 verify.py --replay`	regenerates all seven canonical JSON certificates in memory.

The full replay uses only the Python standard library, deterministic integer, rational, polynomial, and finite-field arithmetic, and canonical JSON serialization. It includes independent paths appropriate to each claim: CSS versus direct-Lagrangian shortening, symbolic versus finite projectivity, chord concurrency versus stabilizer ranks, quotient minors versus finite-field elimination, and cycle-cover versus fraction-free transport determinants. The exact files, hashes, domains, expected outputs, and negative scopes are recorded in `supplement/EVIDENCE.md` and `supplement/EVIDENCE-MANIFEST.json`.

The trusted boundary is Python integer arithmetic, the exact-arithmetic implementations in the generators, deterministic Gaussian elimination, and the mathematical dictionaries connecting arcs, codes, stabilizers, and tensor contractions. The certificates check the bounded enumerations and symbolic identities that they record. They do not prove the conceptual correspondence between the implemented objects and the paper definitions; that correspondence is supplied in the preceding sections.

The accompanying Lean 4 artifact uses toolchain `v4.32.0-rc1`. Its aggregate terminal `RelativeConicArcs.Gates.AMELUAggregate` checks the complete arc–MDS–CSS–AME dictionary, the algebraic pencil quotient, the finite marginal graph counts, the contraction normalization and party-orbit identities, the length-generic LU-to-LC rigidity chain, and the transport polynomial and characteristic-seven identities in one environment. The length-generic part defines the code, equal-phase state, party action, and LU/LC relations on $\mathrm{Fin}(2m)$; proves that the exact $[2m, m, m+1]$ parameters are equivalent to bijectivity of every m-coordinate projection; proves the same MDS property for the dual and constructs full-support $(m+1)$-coordinate shortening generators; and proves the arity-independent full-basis diagonal-flattening axis lemma. The formal proof then expands every shortened $(m+1)$-party marginal on all q^2 Weyl axes, including the identity, proves covariance under the displayed product action, and shows that invertible product maps preserve and reflect the pure contraction locus of a full diagonal tensor. Since $m \geq 2$, two factors remain after contraction, so the locus recovers every local Weyl axis. Retained sets cover all $2m$ parties, and the party permutation is absorbed into the source code. Thus the generic LU-to-LC terminal proves Theorem 1.1 unconditionally for every finite field and every $m \geq 2$. Its $m = 3$ definitions and terminal are reconciled with the established six-party API. Composition with the admitted-pencil interface gives the formal conditional statement LU $\iff z(t) = z(u)$. The same artifact constructs the one-site Clifford quotient by nonzero scalars, proves the scalar subgroup closed, and proves the quotient Hausdorff, discrete, and finite by recording exact Weyl conjugates and constraining their scalar factors through determinant polynomials. It embeds the product-automorphism quotient into a finite product of these one-site quotients. The exact adjoint-signature map is continuous into a finite Hausdorff space, while its identity fiber is the connected product of one-site unit circles. This proves the scalar-phase identity-component statement, both with fixed party labels and after adjoining the discrete party-permutation factor. The scalar inclusions are continuous homomorphisms, every connected component is a scalar-torus translate, and finitely many such components cover either automorphism group. The party-permutation extension has a section-free outer action and a normalized nonabelian factor set. Its ordered change-of-section law makes trivializability independent of the chosen lifts, and trivializability is equivalent to a homomorphic splitting.

The geometric classification, marginal-trace, logical propagation, finite contraction, diagonal-isodual group, and transport orbit steps enter separate hypothesis structures; the paper constructs or certificate-supports them as specified above, but the Lean modules do not. Importing the terminal therefore does not make them unconditional formal theorems. The Choi inverse-transpose bridge, Clifford transpose closure, and factorwise transversal no-go are kernel checked in `EncoderTransversal`. The companion `DiagonalIsoduality` module proves the arbitrary-length MDS diagonal-multiplier lemma, the zero-or-one-dimensional multiplier space, the exact nullity test, and projective uniqueness of the witness ratios. The exact carrier dichotomy remains conditional: special-linearity, torus propagation, diagonal-duality propagation, the off-diagonal block-to-multiplier bridge, and the complete translation fiber remain named fields. Section 3 proves these action-level inputs for arbitrary $[2m, m, m+1]$ MDS codes. The fixed-copy generic-constancy proposition, the party-moving logical normalizer, and the determinant and double-coset inputs of the complete transport theorem are not formalized. The companion `AMELUAggregateAxioms` terminal separates the three native finite graph counts from the ordinary kernel-checked declarations. For each conditional structure, the statement-adequacy ledger records its fields individually. Some fields are constructed in the displayed paper proof, such as the four deck projectivities and the marginal trace/rank reduction; some are mathematical implications supported by a paper-local certificate, such as the H3 determinant count and the four-copy rank evaluations; and some combine a prose construction with a certificate replay, such as the transport determinant and double-coset inputs. These are hypotheses of formalized implications, not additional Lean theorems.

The LU-to-LC mechanism has direct qubit ancestry. Rains recovered the three Pauli axes of distance-two code projectors [Rai99]; Van den Nest, Dehaene, and De Moor used minimal stabilizer supports to force every local factor of an LU equivalence to be Clifford [VdNDDM05]. General minimal-support AME equivalence permits behavior outside our equal-phase CSS hypotheses [BR20, RL25]. The contribution claimed here is the full q^2-axis tensor (with the identity axis fixed by conjugation), its prime-power-qudit extension, and its uniform $[2m, m, m+1]_q$ MDS/CSS specialization. Its operational consequences are factorwise rigidity of transversal code conversions, the non-Clifford no-go, and the exact full-Clifford transversal group of the odd-prime GRS tower. It is not a new LU–LC mechanism in the abstract.

Finally, none of the results classifies nonlinear orthogonal arrays, arbitrary minimal-support AME tensors, non-MDS stabilizer states, or all stabilizer states over prime-power dimensions. We do not claim that four copies are globally minimal, that $z = 2, 4/9$ are orbit exceptions, or that a fixed copy degree separates all classes as q varies. The logical-group theorem and the quantum part of the pencil classification are restricted to odd prime fields. Over extension fields, Frobenius basis permutations are additional local Cliffords and the scalar z is only Frobenius-covariant. The H3 theorem is restricted to the stated odd residue-field reductions, and the detector exceptions described in Appendix A do not alter the characteristic-uniform rigidity theorem.

The central distinction is therefore between covariant and scalar data. A four-party marginal retains the full configuration of local Weyl axes, and that configuration rigidifies every product-unitary intertwiner. Fixed-copy scalar contractions discard those axes and are generically constant, although their rank-jump strata still reveal exact divisors. For the admitted pencil, the covariant rigidity returns the classical degree-eight quotient z as the complete LU coordinate.

A Transport calculation for the four-copy witness

We now explain the rank jump behind (6.4). Fix the six copy permutations

$$\boldsymbol{\sigma} = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

Let V be the three-dimensional message space and let $g_i : V \to k$ be the six coordinate covectors. Form a bipartite four-sheeted cover with vertices L_a, R_b, $0 \le a, b < 4$, and a color-i edge from L_a to $R_{\sigma_i(a)}$. Put V at each vertex and k at each edge, using $g_{p(i)}$ as both restriction maps for a party assignment $p \in S_6$. A global section is a tuple $(x_a, y_b) \in V^8$ satisfying

$$g_{p(i)}(x_a - y_{\sigma_i(a)}) = 0 \tag{A.1}$$

on all 24 edges. The three-dimensional constant-section space is always present. Gauging it away leaves the 24-by-21 matching matrix in (6.3).

Let $U = k^4/k(1,1,1,1)$, and let ρ_i be the action of σ_i on U. For each p, the MDS property makes $g_{p(3)}, g_{p(4)}, g_{p(5)}$ a basis of V^*. Define Q_p by

$$g_{p(i)} = \sum_{k=0}^{2} (Q_p)_{ik} g_{p(3+k)} \qquad (0 \le i < 3). \tag{A.2}$$

Thus Q_p is the left block obtained by systematizing the party-reordered generator matrix on its last three columns; no pivot can vanish because every three coordinate covectors are independent. Put $\widetilde{Q}_p = 2(t-1)Q_p$. This scalar is a unit on the admitted locus. For the identity assignment, the polynomial representative is

$$\widetilde{Q}_1 = \begin{pmatrix} a & b & c \\ a & c & b \\ d & d & d \end{pmatrix}, \qquad \begin{array}{ll} a = -2(t-1)^2, & b = -(t^2-t+1), \\ c = -(t^2-3t+1), & d = -2(t-1). \end{array}$$

If $Y_k \in U$ are the three free copy vectors, then (A.1) reduces to the 9-by-9 block transport operator

$$\widetilde{T}_p = \big((\widetilde{Q}_p)_{ik}(\rho_i - \rho_{3+k})\big)_{i,k=0}^{2}. \tag{A.3}$$

The dimension identity

$$21 - \operatorname{rank} M_{\pi\boldsymbol{\sigma}}(G(t)) = 9 - \operatorname{rank} \widetilde{T}_p \tag{A.4}$$

holds over the field k, where $p = \pi$ in the convention of Section 6. It follows by quotienting the three-dimensional constant-section kernel and then eliminating the three free systematic coordinates. Multiplying the resulting operator by the unit $2(t-1)$ does not change its rank. All steps are invertible on the admitted six-arc locus.

Theorem A.1 (Transport divisor)**.** For a parameter on the admitted odd non-GRS pencil, there exists a party assignment $p \in S_6$ for which the sheaf (A.1) has a nonconstant section if and only if

$$(B^2 - 2A^2)(9B^2 - 4A^2) = 0,$$

or equivalently

$$(z-2)(9z-4) = 0. \tag{A.5}$$

At a generic point above $z = 2$, exactly 96 party assignments have rank 20; at a generic point above $z = 4/9$, exactly 192 do. Every other assignment has rank 21.

Proof. For the identity systematic chart, the coefficients satisfy

$$a = b + c, \qquad d^2 = -2a, \qquad A = a(c-b), \qquad B = bc.$$

Common copy relabelling, a change of basis in V, and rescaling a coordinate covector act on $\widetilde{T}_p$ by invertible row and column operations. Modulo these operations, the 720 party assignments split into boundary types and three nonboundary relative-cover types. Here the type records the six relative copy permutations $\rho_j^{-1}\rho_i$, equivalently the even-cycle pattern in each pair of colored matchings. Expanding a representative determinant by signed cycle covers gives

$$\begin{aligned} P_- &= AB(3B - 2A), \\ P_+ &= AB(3B + 2A), \\ P_2 &= A(B^2 - 2A^2). \end{aligned}$$

With integral bases the determinants are respectively

$$64d^3AB(3B-2A), \quad 64d^3AB(3B+2A), \quad 64d^3A(B^2-2A^2).$$

All omitted factors are units on the admitted odd pencil. Every assignment outside these three types has a determinant that is a unit there. The signed cycle-cover ledger, the exhaustion of the 720 assignments, and a fraction-free determinant path are recorded in the paper-local certificate. The two signed types lie above $w = B/A = \pm 2/3$, hence $z = 4/9$; the axial type lies above $z = 2$. Fraction-free elimination also gives a nonzero 8-minor on each component, so the generic kernel is exactly one-dimensional.

The row/column equivalences organize the exceptional assignments into double cosets. The axial stabilizers have orders 48 and 16, with intersection of order 8, so its support has size

$$48 \cdot 16/8 = 96.$$

For either signed type the corresponding orders are 24, 48, and 6, giving

$$24 \cdot 48/6 = 192.$$

The two signed types are distinguished by the sign of w, while the axial cycle pattern differs from both; hence their support sets are disjoint. Explicit representatives, stabilizer intersections, and complete support sets are in the certificate. Equation (A.4) turns the one-dimensional transport kernels into rank 20 of the matching matrix. $\square$

The exceptional arithmetic belongs to this detector, not to Theorem 1.1. The identity $4B^2 + A^2 = 4G^2$ gives, respectively,

char k	specialized identities
3	$B^2 - 2A^2 = G^2, \quad 9B^2 - 4A^2 = -A^2,$
5	$9B^2 - 4A^2 = 4G^2,$
7	$9B^2 - 4A^2 = 2(B^2 - 2A^2).$

Thus in characteristic 3 both admitted components disappear; in characteristic 5 the $z = 4/9$ component lies on the excluded GRS boundary; and in characteristic 7 the two components merge. The disjoint 96- and 192-element supports then give the exact histogram $20^{288}21^{432}$. Characteristics $11, 13, 41$ introduce only pullback ramification, with no further rank drop. These reductions and the nonzero 8-minors that control them are replayed in the certificate.

The divisor records where one scalar contraction changes rank. It is not an LU-orbit boundary and, by Proposition 6.3, cannot be promoted to a generic coordinate.

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